

SALESFORCE DEVELOPER PROJECT

Salesforce CRM Application for Banquet Hall Booking

A CRM system to streamline venue bookings, vendor coordination,
event scheduling, and payment tracking

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1. Project Overview

Project Overview: A CRM Application for Banquet Hall Booking

Objective

This project develops a comprehensive booking management system for a banquet hall business using Salesforce. The system streamlines the booking process, manages event details, and provides insights into sales and revenue. Additionally, it includes features to manage catering/menu selection, DJ services, decorations, themes related to weddings, and other marriage-related services. Flows are implemented to ensure data integrity and prevent double bookings.

2. Ideation Phase

The Ideation Phase serves as the foundation of any successful project. It blends creativity, user empathy, and structured thinking to identify the core challenges, generate meaningful ideas, and prioritize solutions that bring tangible value to all users involved. This dynamic phase combines imagination with insight to ensure every feature addresses real-world needs.

For our project, **“A CRM Application for Banquet Hall Booking,”** the ideation process was essential to tailor the system to the specific needs of venue managers, event organizers, service vendors, and customers. By deeply understanding their workflows and challenges, we focused on creating a solution that streamlines bookings, improves client communication, and enhances overall event planning efficiency.

2.1 Brainstorming & Idea Prioritization Template

Step 1: Team Gathering, Collaboration, and Selecting the Problem Statement

Our team convened with the goal of identifying inefficiencies in the current banquet hall booking process and proposing a tech-enabled solution using Salesforce. Through collaborative brainstorming sessions, digital whiteboards, and real-world scenario mapping, we collectively analyzed the operational challenges faced by banquet hall managers, event organizers, and support staff. We examined existing workflows and discovered that most banquet halls still rely on fragmented and manual systems for managing:

- Booking and scheduling of halls
- Customer communications and inquiries
- Payment tracking and invoice generation
- Event coordination and vendor assignment

Step 2: Brainstorm, Idea Listing, and Grouping

We conducted a collaborative brainstorming session using a digital whiteboard where each team member contributed raw ideas based on industry research and user needs. The ideas were then reviewed and grouped into major themes.

Step 3: Idea Prioritization

Top Priority Features:

- Real-time booking calendar with conflict prevention
- Customer profile management and event history tracking
- Automated reminders and confirmations (via SMS/Email/WhatsApp)
- Interactive dashboards showing hall utilization, revenue, and peak demand times
- Secure, role-based access for admins, coordinators, and vendors

2.2 Empathy Mapping – Empathize & Discover

An **empathy map** is a visual tool that helps teams develop a deeper understanding of users' experiences, frustrations, and aspirations. We used this tool to map the daily journey of key banquet hall booking stakeholders—including **venue managers, event coordinators, vendors, and customers**—to surface the emotional and operational realities they face.

Empathy Map Canvas

SAY	THINK	DO	FEEL
<i>"We're overloaded with calls and double bookings."</i>	<i>"A single dashboard for all bookings would help."</i>	<i>"Do we have all the resources for the event?"</i>	<i>"I want a stress-free and smooth booking process."</i>
Venue Managers	Event Organizers	Vendors / Staff	Clients / Guests
• Manually checks diaries or spreadsheets • Anxious, uncertain, time-constrained	• Call for follow-ups and availability • Communicate on WhatsApp, make calls	• Communicate on WhatsApp, make calls • Visit venue, send inquiries through various apps	• Stressed, reactive, under pressure • Confused, doubtful, hopeful

Figure 1: Empathy map of banquet hall booking stakeholders

By stepping into the user's shoes, we ensured that our Salesforce CRM features (formulas, flows, triggers, dashboards) directly addressed their key frustrations.

2.3 Define the Problem Statements

Customer Problem Statement Template

Banquet hall operators and event managers currently manage bookings, customer interactions, and vendor coordination through fragmented and often manual processes. This results in scheduling conflicts, missed follow-ups, payment tracking errors, and lack of visibility into overall business performance. A centralized Salesforce CRM system can streamline booking workflows, manage customer data efficiently, automate communications, and provide real-time insights through dashboards, flows, and automation. This statement reflects the expectations of venue managers and clients and served as a guiding reference for our object models, data validations, process automations, and reporting configurations.

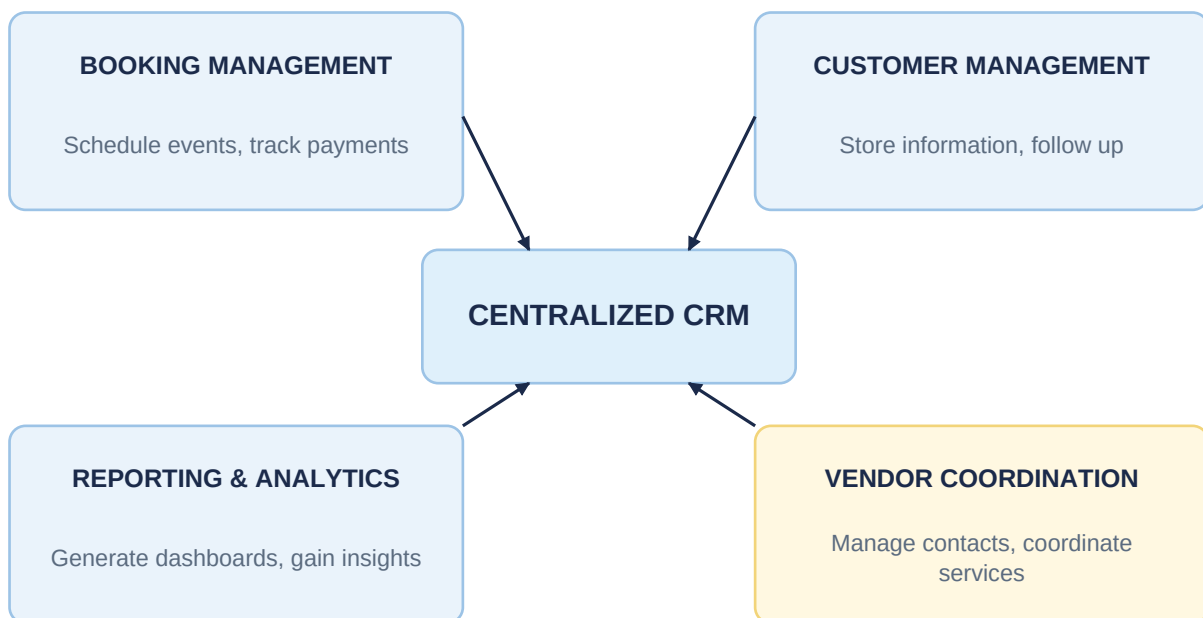


Figure 2: Centralized CRM for Banquet Hall Booking

3. Requirement Analysis

In our project, “A CRM Application for Banquet Hall Booking,” this phase acted as the bridge between identifying booking-related challenges and shaping an intelligent system design. Through detailed user journeys, workflow analysis, and functional requirement gathering, we outlined critical needs such as real-time scheduling, customer data handling, vendor assignment, and seamless communication.

3.1 Customer Journey Steps

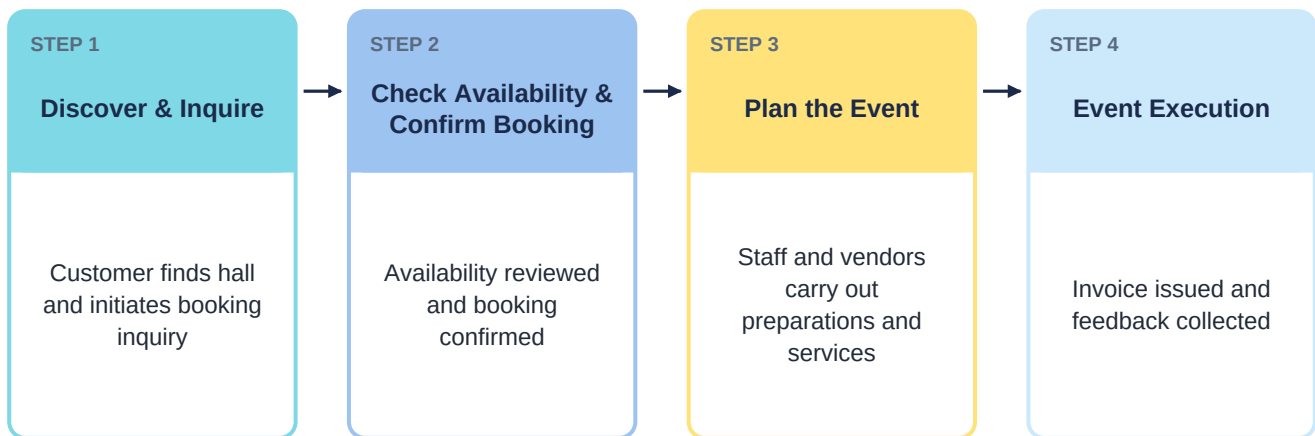


Figure 3: Customer journey from inquiry to event execution

3.2 Data Flow Diagram

Purpose: Mapping information flow between objects.

The **Data Flow Diagram (DFD)** models how information moves between Salesforce objects and components in the banquet hall CRM system. It helped us structure relationships between **Customer**, **Booking**, **Venue**, **Event Details**, **Vendor**, and **Payment**.

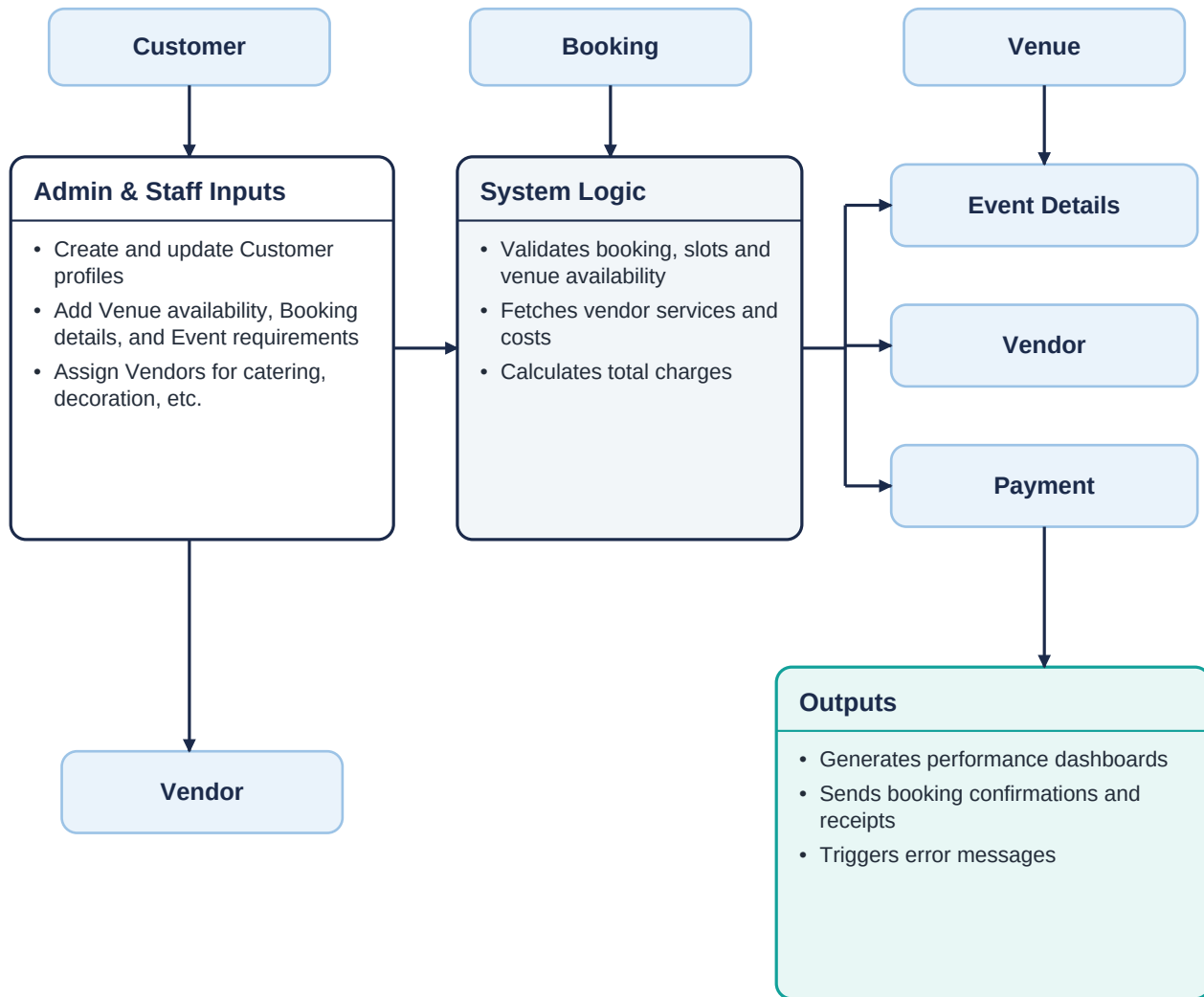


Figure 4: Data flow between Customer, Booking, Venue, Event Details, Vendor, and Payment

3.3 Solution Requirements

What the CRM system must do – Functional Requirements:

- **Create custom objects for:**
 - **Venue** (with capacity, location, availability)
 - **Booking** (with date, time, customer, event type)
 - **Customer** (contact info, history, preferences)
 - **Vendor** (services, costs, availability)
 - **Payment** (methods, totals, transaction status)
- **Establish relationships** using **lookup**, **master-detail**, and **formula fields**:
 - Booking → Customer, Venue, Vendor

- Payment → Booking
- Dynamic pricing logic via formulas (e.g., duration × rate)

CRM Technologies Used

Category	Technology Used	Description
Platform	Salesforce Lightning	Used to build custom CRM using standard & custom objects
Automation	Flows & Validation Rules	Automate booking confirmations and enforce data accuracy
Custom Logic	Apex Triggers & Classes	Used for custom business logic and validations
Reports & Dashboards	Salesforce Reports	To analyze booking data, revenue, vendor usage, etc.
UI/UX	Lightning App Builder	For creating a unified app view for Banquet Hall Booking

4. Project Design Phase

The Project Design Phase laid the technical and functional blueprint for the banquet hall CRM system, ensuring it addressed real user needs while remaining scalable and aligned with Salesforce best practices. It transformed validated pain points into structured objects, flows, and business rules. This phase guided decisions around data relationships, automation logic, and UI components, bridging the gap between user insights and a buildable, maintainable solution.

4.1 Problem–Solution Fit

Does the Proposed Solution Fit?

Yes. The Salesforce CRM solution:

- Introduces object-level control for **Bookings, Venues, Customers, Vendors, and Payments**
- Enables automated **service pricing and availability checks** through Flows
- Validates **vendor assignments and booking conflicts** using Apex Triggers
- Summarizes operational performance using **Reports and Dashboards**
- Uses **formula fields** to dynamically calculate charges and event costs
- Creates **centralized scheduling and coordination views** using Lightning App Builder

Thus, it directly addresses the operational pain points of managing banquet hall bookings and event logistics in a unified, scalable, and efficient manner.

Pros and Cons

PROS	CONS
Centralized Booking Management All bookings, customer details, and vendor assignments are stored in one place—reducing errors and improving coordination.	High Initial Setup Cost Customizing Salesforce for banquet operations may require developer time and licensing costs.
Automation of Routine Tasks Flows and triggers can automate confirmations, reminders, and payment alerts, saving time and reducing manual follow-ups.	Learning Curve Staff may need training to navigate the Lightning interface and understand CRM workflows.
Customizable to Business Needs Salesforce allows you to tailor objects, fields, and logic to match your specific banquet operations and workflows.	Overhead for Small Venues For smaller banquet halls with limited bookings, the system might feel too complex or underutilized.
	Ongoing Maintenance Custom flows, triggers, and reports need periodic updates to stay aligned with business changes.

4.2 Proposed Solution

Our proposed CRM application is designed to digitize and streamline banquet hall booking operations using Salesforce's powerful customization capabilities.

Key Functional Features

- **Custom Objects:** Venue, Booking, Customer, Vendor, Event Details, and Payment
- **Automation & Validation:**
 - Vendor assignment checks (e.g., caterers, decorators) using Apex Triggers
 - Total cost calculation automation using Flows based on services and duration
 - Input validation for booking conflicts, missing information, or overbooking

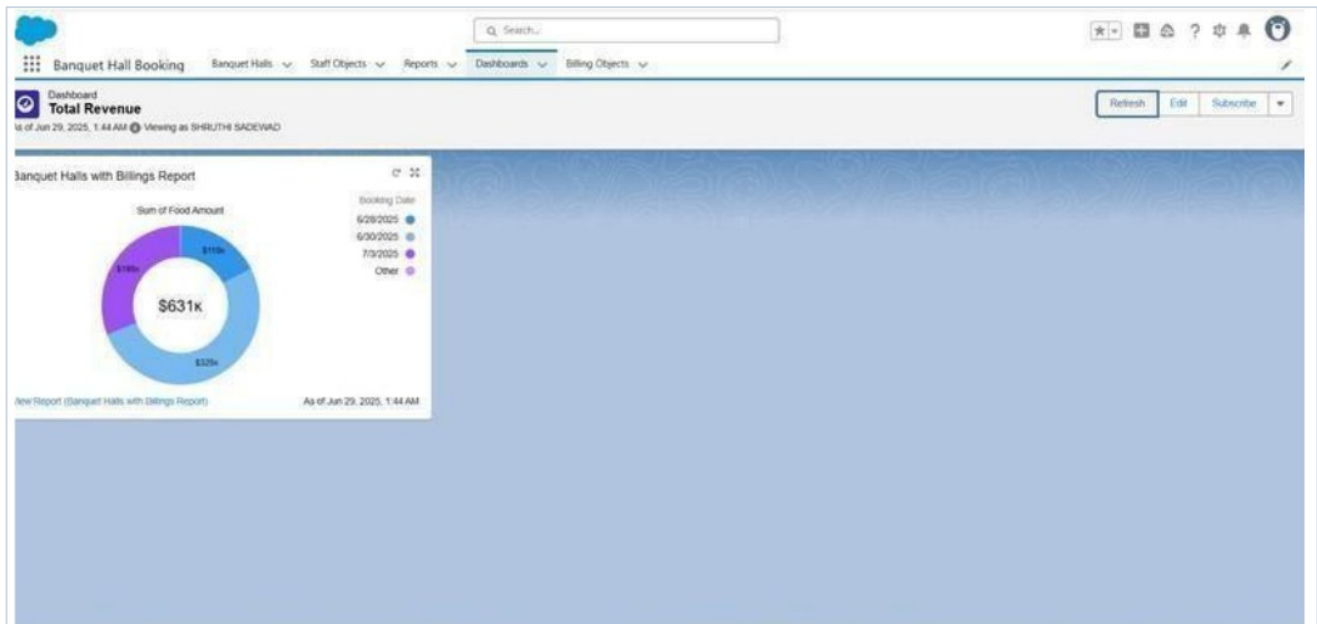


Figure 5: Total Revenue dashboard (Banquet Halls with Billings report)

4.3 Solution Architecture

Object Relationship Overview

Object	Key Fields / Features
Venue	Name, Location, Capacity, Availability
Booking	Linked to Venue, Start Date, End Date, Status
Customer	Name, Email, Phone, Address
Vendor	Name, Service (Picklist), Phone, Email
Event Details	Linked to Customer and Vendor, Number of Guests, Setup Details Formula

Lookup Relationships

- Staff Member → Venue
- Booking → Venue, Customer, Event Coordinator, Vendor Package
- Vendor Package → Vendor

5. Project Executable Files

This phase outlines the actual Salesforce configurations, data models, and outcomes implemented during the execution of the Banquet Hall Booking CRM application. It ensures that all key elements—custom objects, automation flows, validations, and reports—are clearly documented and aligned with real-world use cases. The components developed in this phase are designed to be traceable, reusable, and easily assessable for future improvements or audits. It captures the practical realization of booking flows, vendor assignment rules, and payment tracking modules, offering clarity, replication, and validation across the entire system lifecycle.

5.1 Project Files

The following project files were executed in the Salesforce Developer Org:

Milestone	Description
Milestone 1	Developer Account Setup
Milestone 2	Object Creation
Milestone 3	Tab Creation
Milestone 4	Lightning App Setup
Milestone 5	Field Creation
Milestone 6	Page Layouts
Milestone 7	Creation of Record Types
Milestone 8	Validation Rules
Milestone 9	Flow Setup
Milestone 10	Apex Trigger
Milestone 11	Reports
Milestone 12	Dashboards
Milestone 13	Final Review

6. Output Screenshots

Milestone 1: Developer Account Setup

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Figure 6: Salesforce Developer Edition sign-up

Milestone 2: Object Creation

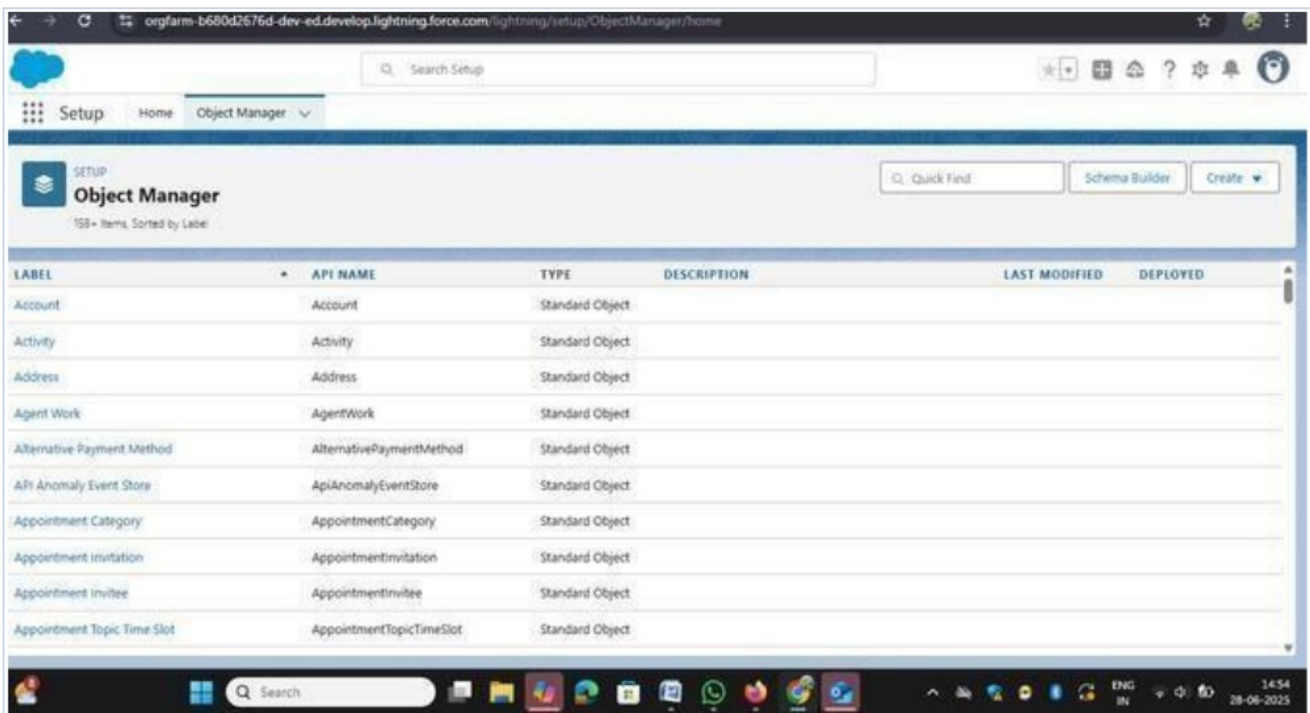


Figure 7: Object Manager listing objects in the Developer Org

Milestone 3: Tab Creation

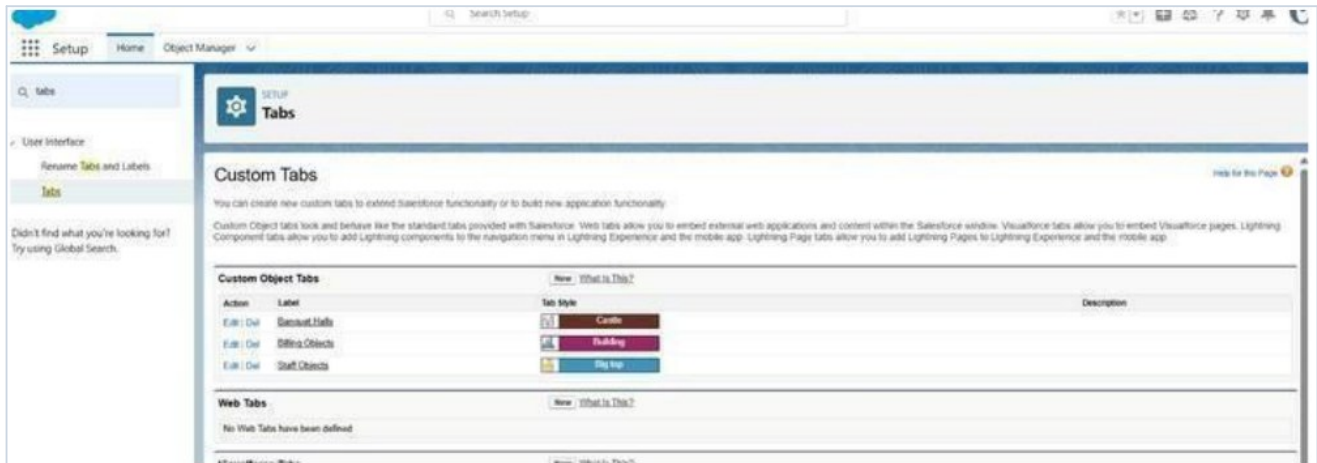


Figure 8: Custom tabs created for the application

Milestone 4: Lightning App Setup

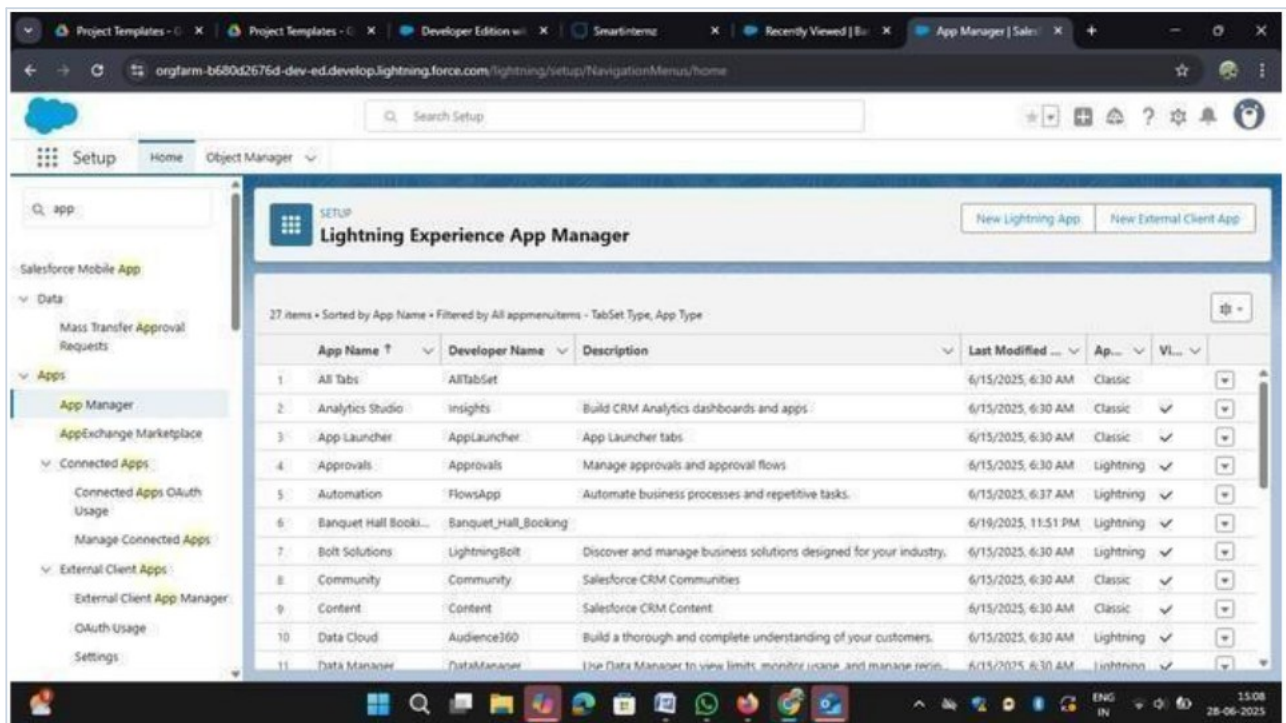


Figure 9: Lightning Experience App Manager showing the Banquet Hall Booking app

Milestone 5: Field Creation

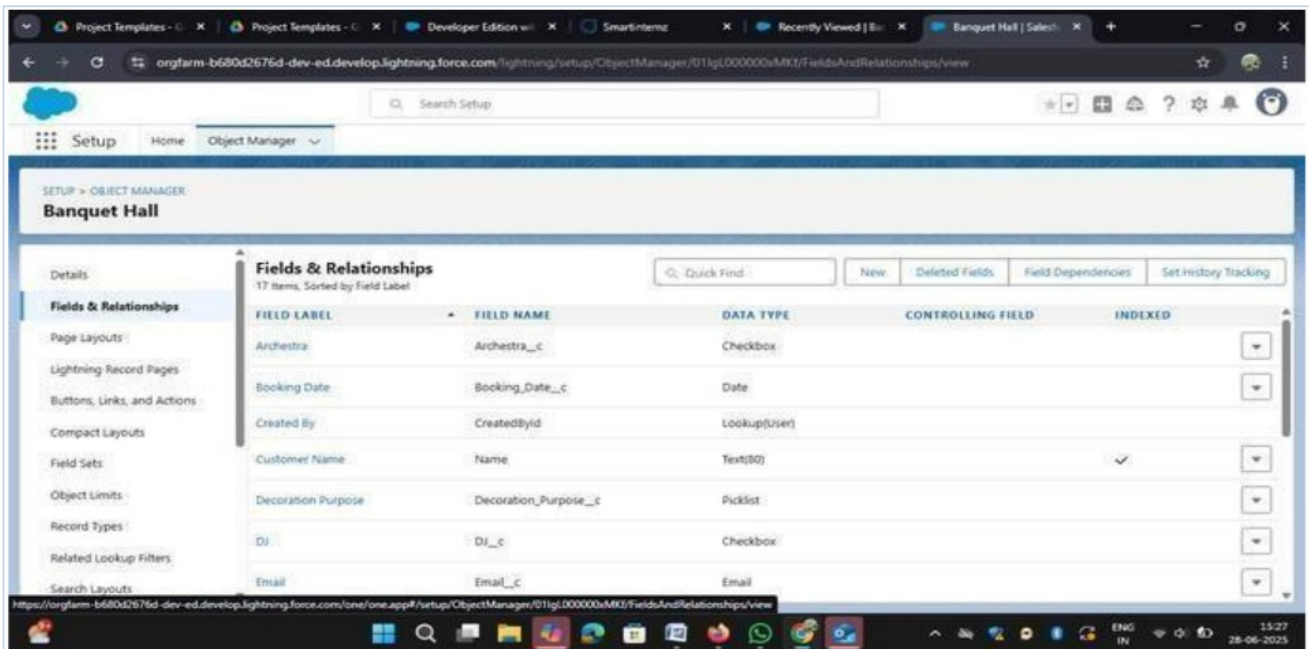


Figure 10: Banquet Hall object – fields and relationships

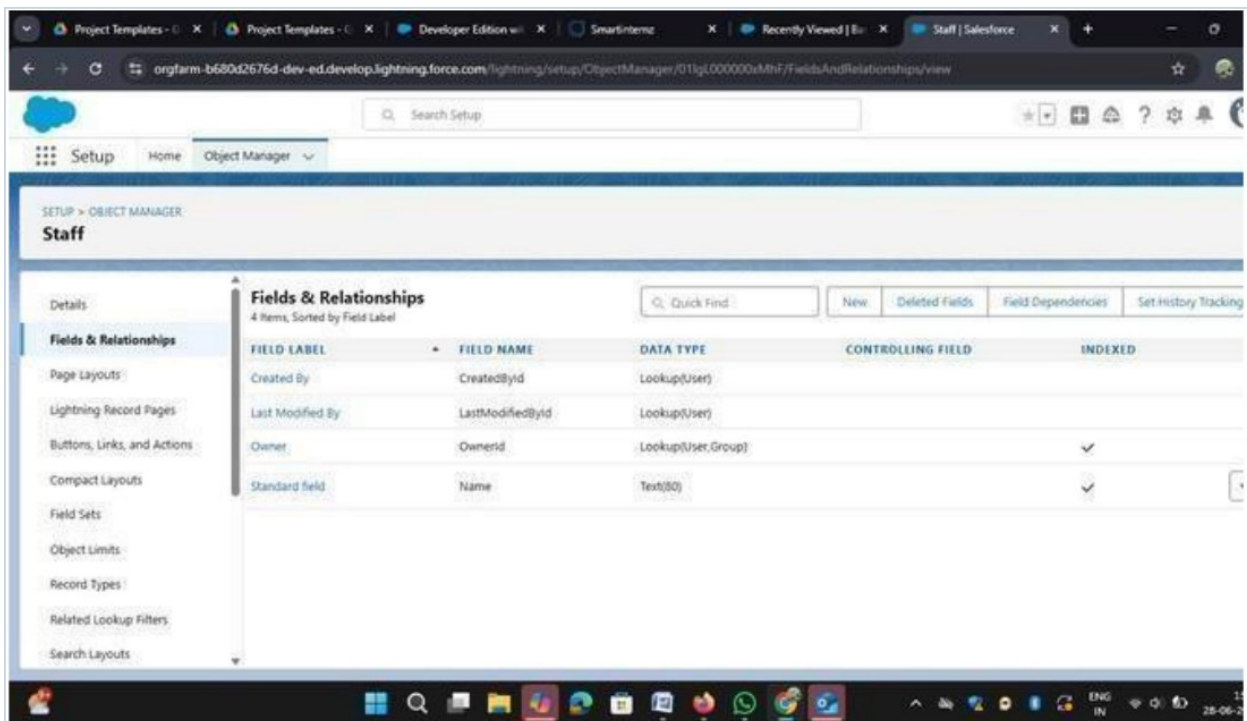


Figure 11: Staff object fields

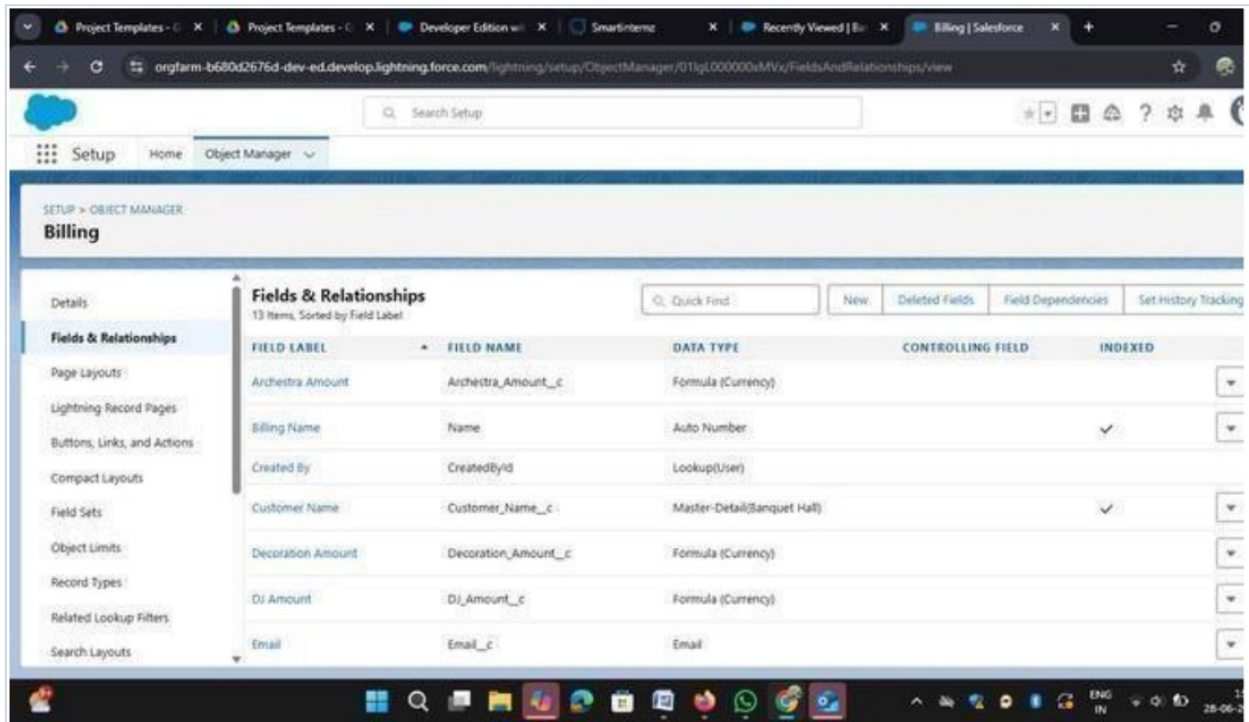
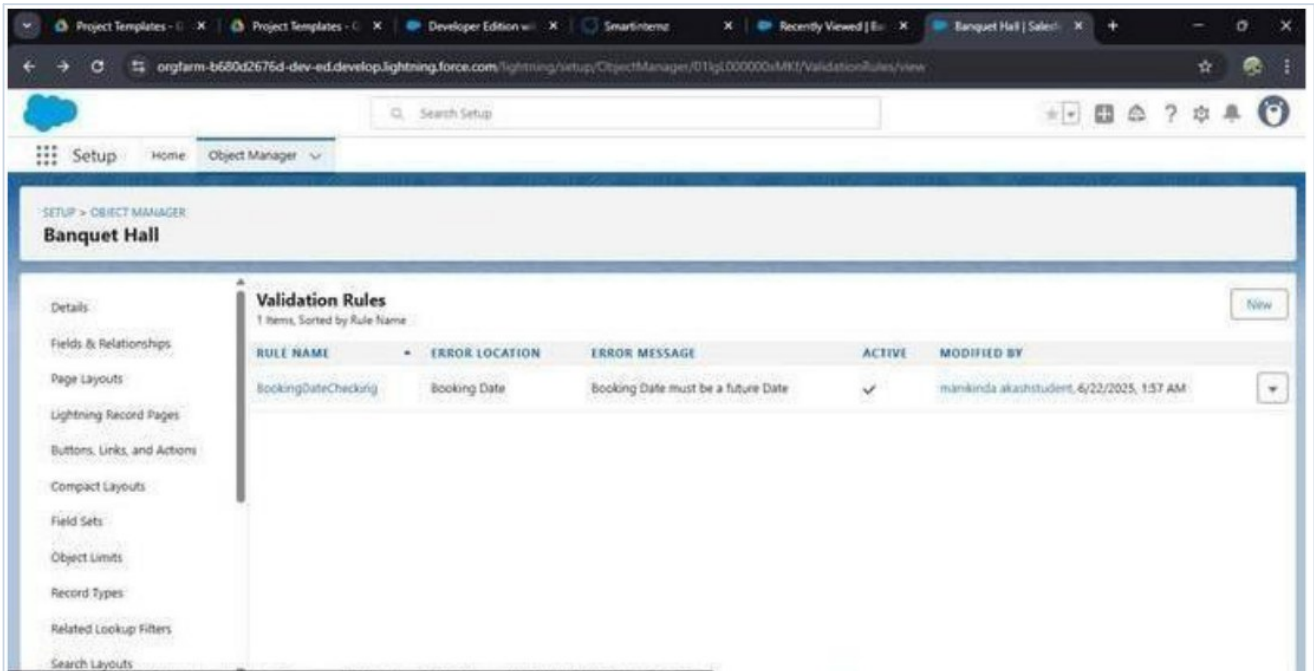


Figure 12: Billing object fields

Milestone 6: Page Layouts

- Customized page layouts per object.
- Grouped fields logically for usability.

Milestone 8: Validation Rules



The screenshot displays the Salesforce Setup interface for the 'Banquet Hall' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, and Search Layouts. The main content area is titled 'Validation Rules' and shows a table with one rule.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
BookingDateChecking	Booking Date	Booking Date must be a future Date	✓	mankinda akashstudent, 6/22/2025, 1:57 AM

Figure 13: Validation rule *BookingDateChecking* on the *Banquet Hall* object

Milestone 9: Flow Setup

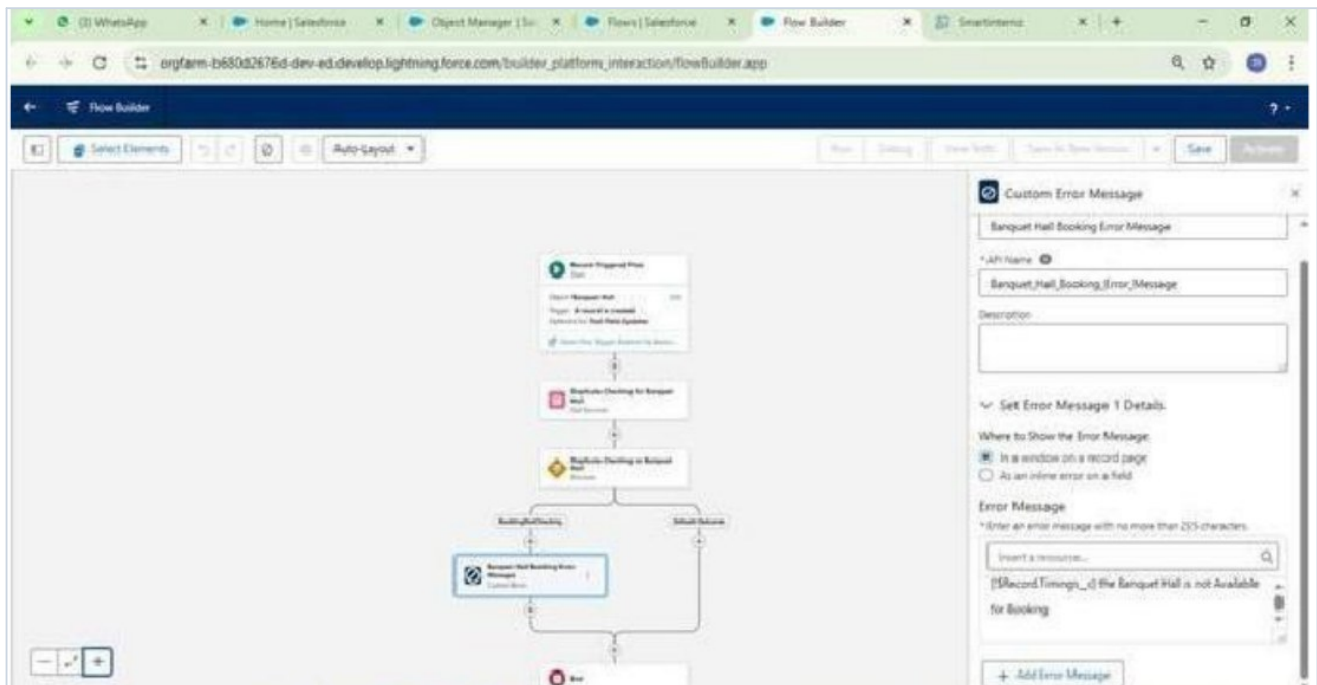


Figure 14: Record-triggered flow with a custom error message when the hall is not available

Milestone 10: Apex Trigger

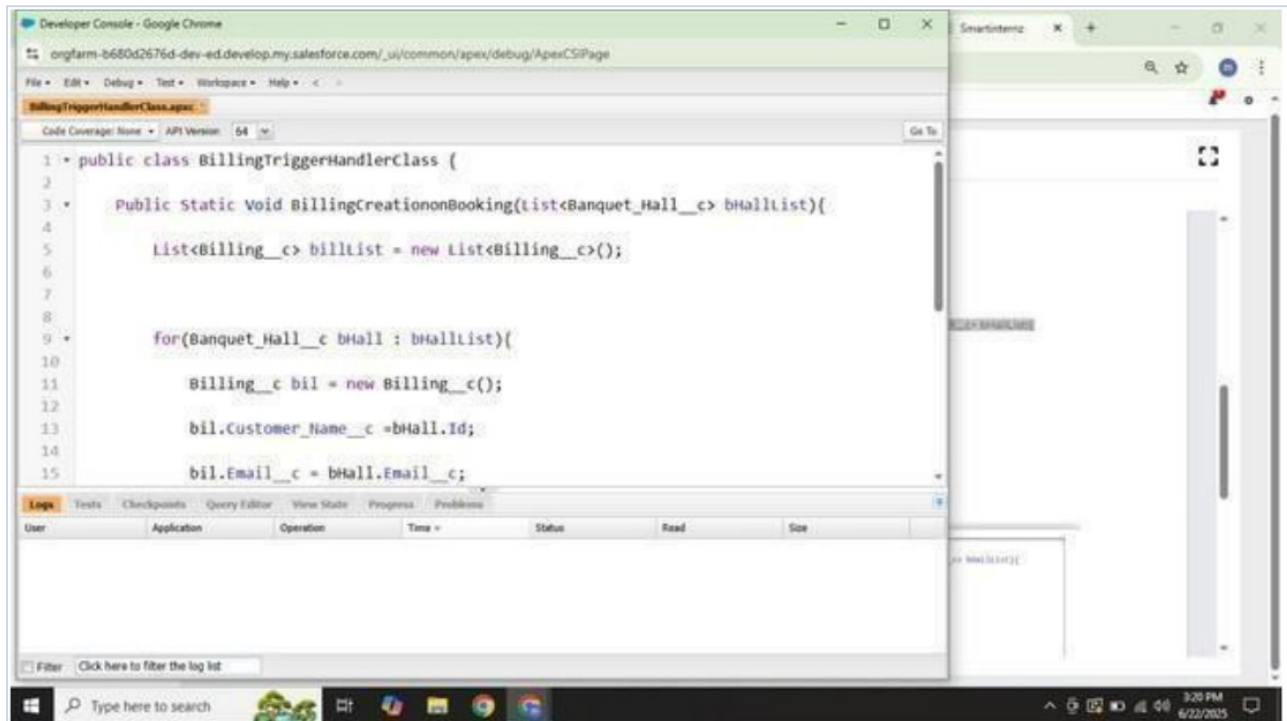


Figure 15: BillingTriggerHandlerClass in the Developer Console

7. Results

Key functional screenshots include:

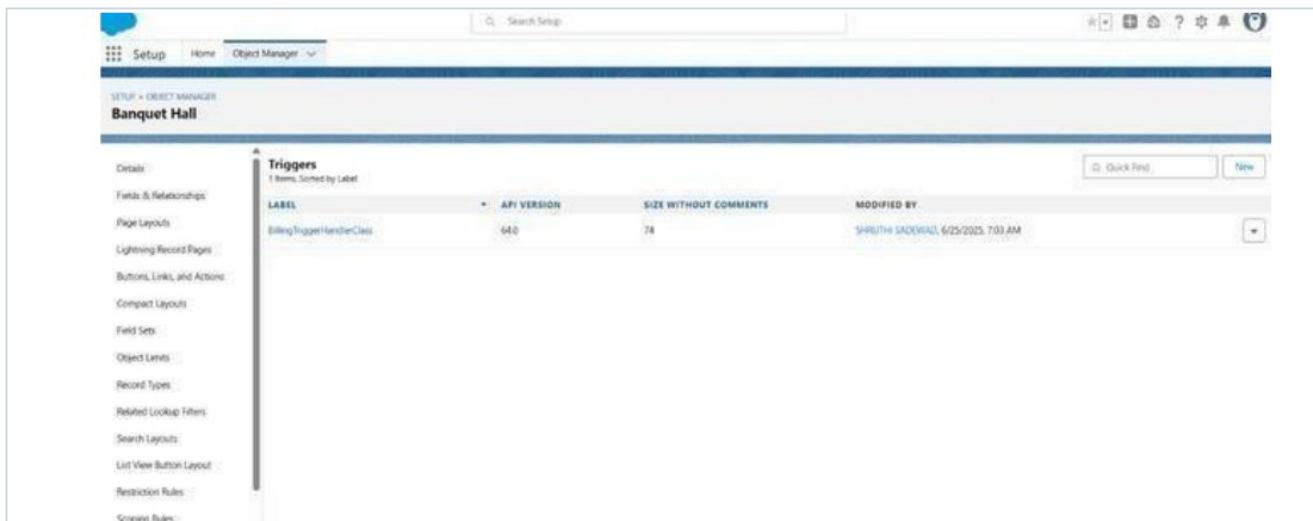


Figure 16: Apex trigger listed on the Banquet Hall object

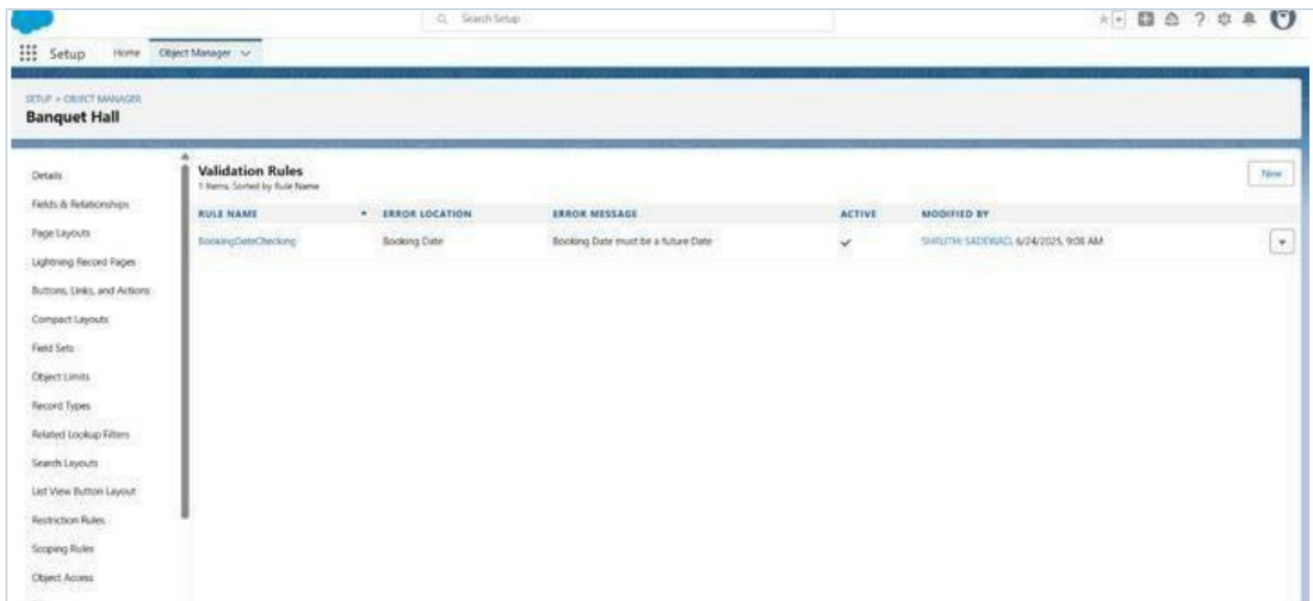


Figure 17: Active validation rule on the Banquet Hall object

8. Advantages & Disadvantages

Advantages	Disadvantages
- Real-time data validation and automation - Modular structure allows easy customization - Centralized and streamlined process for booking, event cost, and staff coordination - Visual reports and dashboards for decision-making	- Requires knowledge of Salesforce for advanced use - Limited functionality in offline mode due to cloud dependency - Dependent on correctly established object relationships

9. Conclusion

The project successfully delivered a customized Salesforce CRM application tailored to banquet hall booking operations. It streamlined event management workflows, addressed key operational inefficiencies, and enhanced user coordination across venues, vendors, and customers. The initiative also offered valuable learning opportunities for the team in both declarative and programmatic Salesforce development, reinforcing best practices in CRM design, automation, and scalability.

RESULT

Thus the above Salesforce project of CRM application for banquet hall booking has been completed successfully.